

# AJAYKRISHNAN E S

Email : [aj66@ucsb.edu](mailto:aj66@ucsb.edu)

Website: [ajaykrishnan-es.github.io](https://ajaykrishnan-es.github.io)

## EDUCATION

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| <b>Ph.D. in Computer Science</b><br>University of California, Santa Barbara<br>Advisor: Prof. Daniel Lokshtanov<br>Topics: Graph Algorithms, Algorithmic Graph Theory              | 2023 - current<br>CGPA - 4/4 |
| <b>BS-MS (Specialisation in Mathematics)</b><br>Indian Institute of Science Education and Research, Pune<br>Thesis Title: Exact Exponential Algorithms & KNOT-FREE VERTEX DELETION | 2018 - 2023<br>CGPA - 9.2/10 |

## AWARDS AND RECOGNITION

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| Recipient of <b>Chuck Dana Research Accelerator Award</b> at UC Santa Barbara.<br><i>Awarded \$8,000; sole recipient from a competitive pool of UCSB CS graduate students.</i>               | 2026 |
| Co-author of the paper that received the <b>Mihai Pătraşcu Best Paper Award</b> at SOSA.<br><i>Title: A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs.</i>                    | 2026 |
| Recipient of the <b>Outstanding Teaching Assistant</b> award at UC Santa Barbara.<br><i>Nominated by graduating seniors; awarded to one TA per department annually.</i>                      | 2025 |
| Recipient of UCSB CS <b>Summer Academic Fellowship</b> .<br><i>Awarded \$10,000 to support independent research over the summer.</i>                                                         | 2024 |
| Recipient of the <b>Xytel Best Thesis Award</b> in Mathematics at IISER Pune.<br><i>Awarded to one graduating student per discipline each year.</i>                                          | 2023 |
| <b>Graduated with distinction</b> from IISER Pune.<br><i>An honor reserved for students with a CGPA above 9 (approximately the top 5%).</i>                                                  | 2023 |
| Recipient of the <b>Innovation in Science Pursuit for Inspired Research</b> scholarship.<br><i>Five-year fellowship from the Government of India for pursuing higher studies in science.</i> | 2018 |

## VISITS AND WORKSHOPS

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| <b>SoCal Theory Day</b> at University of California San Diego                                                         | Feb 2026    |
| <b>Institute of Mathematical Sciences, Chennai.</b><br>Host: Prof. Saket Saurabh.<br>Topic: Parameterized Algorithms. | Summer 2025 |
| <b>Workshop on Hardness of Approximation in P</b> at DIMACS, Rutgers University                                       | July 2025   |
| <b>Fusing Theory and Practice of Graph Algorithms</b> at ICERM, Brown University                                      | Feb 2025    |
| <b>Extroverted Sublinear Algorithms</b> at Simons Institute                                                           | June 2024   |
| <b>Randomized Methods for Approximation and Parameterized Algorithms.</b>                                             | Dec 2022    |
| <b>Algorithms - To Improve, or Not to Improve, That's the Question!</b>                                               | Jan 2022    |

## SELECT PUBLICATION

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*Note: In theoretical computer science, authors are listed in alphabetical order. Among my publications at venues which use ranked authorship, those marked with (★) indicate equal contribution from all authors.*

### **Parameterized Approximation of Rectangle Stabbing.**

Huairui Chu, **Ajaykrishnan E S**, Daniel Lokshtanov, Anikait Mundhra,  
Thomas Schiber, Xiaoyang Xu, Jie Xue.

*ESA 2026*

### **A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs. (Best Paper Award)**

**Ajaykrishnan E S**, Robert Ganian, Daniel Lokshtanov, Vaishali Surianarayanan.

*SOSA 2026*

### **Fast Hypertree Decompositions via Linear Programming: Fractional and Generalized. (★)**

Vaishali Surianarayanan, Anikait Mundhra, **Ajaykrishnan E S**, Daniel Lokshtanov.

*SIGMOD 2025*

### **Beyond Exact Fairness: Envy-Free Incomplete Connected Fair Division.**

**Ajaykrishnan E S**, Daniel Lokshtanov.

*FSTTCS 2025*

## UNDER SUBMISSION

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### **Induced-Minor-Closed Classes have Linear, Square-Root, or Sub-Polynomial Tree-Independence.**

*Submitted - SODA*

Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

### **(Treewidth, Clique)-Boundedness and Poly-logarithmic Tree-Independence.**

*Submitted - IJM*

Maria Chudnovsky, **Ajaykrishnan E S**, Daniel Lokshtanov.

### **Induced Minors and Coarse Tree Decompositions.**

*Submitted - JGT*

Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

### **Computing Join Trees for Join Queries with Inequalities.**

*Submitted - PODS*

**Ajaykrishnan E S**, Daniel Lokshtanov, Anikait Mundhra,  
Vaishali Surianarayanan, Nikolaos Tziavelis.

## SERVICE

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### **Conference Reviewing:**

FOCS 2024, ITCS 2025, COCOON 2025, ESA 2025, STACS 2026, SODA 2026, IWOCA 2026, MFCS 2026.

### **Mentorship Roles:**

UC Santa Barbara, Early Research Scholars Program; a research program that seeks to give students, *particularly those who are traditionally under-served* in computing the foundational knowledge and skills for engaging in research in the discipline.

- Mentored a team of three undergraduate students.

2024-2025

*Project Title: Teaching Assistant Assignment.*

- Mentoring a team of four undergraduate students.

2025-2026

*Project Title: Low Distortion Metric Embedding.*

**Teaching Experience:**

- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II  
*With Prof. Subhash Suri; Fall 2023, Spring 2024, Winter 2025.*
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II  
*With Prof. Eric Vigoda; Winter 2024.*
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II  
*With Prof. Daniel Lokshtanov; Fall 2024.*
- Teaching Assistant at UCSB for CMPSC 134 - Randomized Algorithms  
*With Prof. Eric Vigoda; Spring 2025.*
- Teaching Assistant at UCSB for CMPSC 40 - Foundations of Computer Science  
*With Dr. Vaishali Surianarayanan; Summer 2024.*
- Teaching Assistant at IISER-Pune for MTH 323 - Graph Theory  
*With Prof. Soumen Maity; Fall 2022.*